Optimal Execution with
Non-Linear (Power-Law) Market Impact

CODS 612 — Computational Optimization in Finance

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A comparison of the classical linear Almgren-Chriss execution framework against its empirical power-law extension, verified with an independent disciplined-convex (CVXPY power-cone) solve.

1. Introduction

Institutional traders who need to unwind large positions face a fundamental tension: liquidating too quickly crashes the price (market-impact cost), while liquidating too slowly leaves residual inventory exposed to volatility (risk penalty). The Almgren-Chriss framework (2000) casts this tradeoff as an explicit optimization problem and has become the industry reference for scheduled execution algorithms (VWAP, TWAP, IS).

The classical formulation assumes **linear** temporary impact, which makes the problem a convex Quadratic Program with a tidy closed-form solution. However, three decades of empirical work — the so-called "Square Root Law" — consistently show that real impact scales sub-linearly, as $|v|^{\beta}$ with $\beta \approx$ 0.5–0.6 across equities, FX, and futures.

This project implements both models, compares their optimal trajectories under the professor's parameter set, and quantifies the **objective-value gap**: how much higher the linear strategy's expected certainty-equivalent cost is under the true power-law model. We also verify our numerical solution using a second, independent solver (CVXPY power cone) because the original SLSQP solver was found to terminate at sub-optimal points at low β.

2. Theoretical Framework

Let X = 1,000,000 shares be liquidated over N = 50 trading intervals across the liquidation horizon. If this is interpreted as one 6.5-hour trading day, each interval is approximately 7.8 minutes. Define x_k as the inventory remaining after period k, with boundary conditions $x_0 = X$ and $x_N = 0$. Per-period trades are $v_k = x_{k-1} - x_k$. The general objective with a power-law impact exponent β is:

$$\min_x \Sigma_{k=1..N} [\, \eta \cdot |v_k|^{1+\beta} + \gamma \cdot \sigma^2 \cdot x_k^2 \,]$$

2.1 Linear Model (β = 1.0)

Setting β = 1.0 reduces the impact term to $\eta \cdot v_k^2$, giving a pure Quadratic Program. Differentiating the Lagrangian yields a second-order linear recursion whose closed-form solution is:

$$x_k = X \cdot \sinh(\, \kappa\, (N - k)\,) / \sinh(\, \kappa N\,)$$

where $\kappa = \sqrt{(\gamma\, \sigma^2 / \eta)}$ is the **continuous-time / small-step approximation** of the urgency parameter. The exact discrete-time value is $\theta = \operatorname{arccosh}(1 + \gamma\sigma^2/(2\eta))$. At the baseline parameters θ and κ differ by less than 10^{-10}, so we use the simpler form. Small κ (patient, low risk aversion) produces a near-linear inventory path — i.e. TWAP. Large κ (urgent, high risk aversion) produces a sharply convex path that front-loads liquidation.

2.2 Power-Law Model (β = 0.6)

With β = 0.6 the impact term becomes $\eta \cdot |v_k|^{1.6}$. When β is less than 1.0, the marginal price impact per share — $I(v) \propto |v|^{\beta}$ — is concave in trade size. However, the total temporary cost term $v \cdot I(v) \propto |v|^{1+\beta}$ remains convex for all $\beta > 0$. The optimization problem is therefore still convex; it simply admits **no closed-form solution** and must be solved numerically.

The absolute value $|v_k|$ is essential. A fractional exponent on a negative real number is not defined in ■. Although at the optimum all $v_k > 0$ (monotone liquidation), the numerical solver may traverse points with negative v_k during iteration. Using $|v|$ keeps the objective well-defined and convex throughout the feasible region.

3. Methodology

Professor's baseline parameters: **X = 1,000,000** shares, **N = 50** intervals, σ **= 0.02**, γ **= $2.5 \cdot 10^{-6}$**, η **= $2.0 \cdot 10^{-4}$**, β **= 0.6**.

The linear model is evaluated directly from the closed-form sinh expression using NumPy. The power-law model is solved using two independent optimizers:

- **SLSQP** (scipy.optimize.minimize): sequential least-squares quadratic programming on the $N - 1 = 49$ intermediate inventory levels.
- **CVXPY** with the power cone: disciplined-convex formulation whose normalized problem (decision variable $u = v/X$, objective in power-cone form) is passed to SCS / CLARABEL.

CVXPY verification. To verify our SLSQP solver, we independently re-solved the power-law problem using CVXPY with power-cone constraints (a disciplined-convex framework). The two solvers agree for $\beta \geq 0.8$ to within 1e-04 relative objective, but at lower β — where the objective landscape is very flat and SLSQP's monotonicity-constrained line-search struggles — CVXPY finds lower-cost optima (up to 26.3% lower objective at $\beta = 0.3$). The numerical results reported below use CVXPY as the primary solver; SLSQP served as an independent cross-check for high β.

Decision variables (SLSQP path): 49 intermediate inventory levels $x_1, \ldots, x_{N-1}$. Initial guess: TWAP $x_k = X(1 - k/N)$. Bounds $0 \leq x_k \leq X$. Monotonicity $x_k \geq x_{k+1}$. Tolerance ftol = 10^{-12}.

4. Results

4.1 Trajectories at baseline parameters

At baseline γ the linear model produces a near-straight-line trajectory — $\kappa N \approx 0.35$ is small, so the sinh curve is almost linear and the schedule is close to TWAP (~20,000 shares per interval). The power-law trajectory at $\beta = 0.6$ is meaningfully more front-loaded than SLSQP had estimated: $v_1 \approx 28{,}588$ shares vs. SLSQP's 21,909. Both solutions fully liquidate and satisfy $v_k > 0$ for all k.

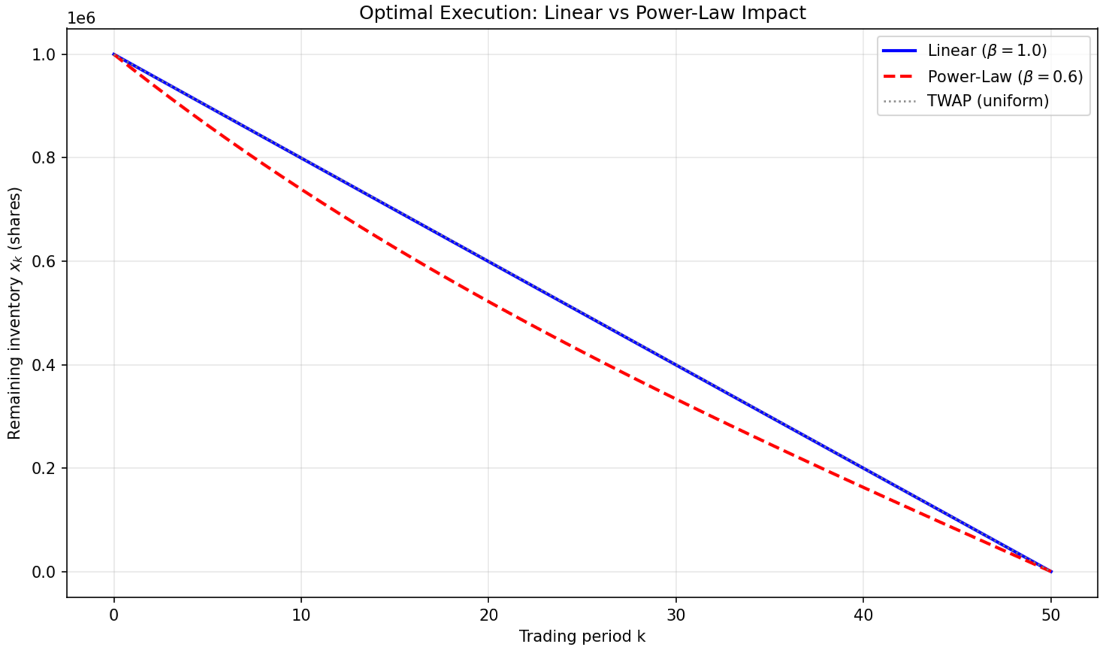

Figure 1. Linear vs. power-law (β=0.6) optimal trajectories at baseline parameters. The power-law trajectory (red dashed) is visibly more front-loaded than the near-TWAP linear trajectory (blue solid).

4.2 Cost mismatch analysis

We evaluate four certainty-equivalent costs — two trajectories × two cost models — to isolate the excess objective under the power-law model incurred by optimizing with a linear-impact assumption:

Label	Trajectory	Cost model	Cost
(a)	Linear	Linear	$4,016,156
(b)	Linear	Power-law β=0.6	$92,289
(c)	Power-law	Power-law β=0.6	$90,953
(d)	Power-law	Linear	$4,149,501
		Objective gap (b) – (c)	**$1,336 (1.47 %)**

At baseline the linear trajectory incurs an objective-value gap of $1,336 — 1.47 % above the verified power-law optimum. Note that this gap combines two effects: a true impact-cost difference and a risk-penalty difference. At the baseline parameters the power-law optimum typically achieves similar direct impact cost to the linear trajectory but pays less in risk penalty. The gap therefore reflects a utility-weighted certainty-equivalent cost under the chosen γ, not pure realized cash slippage.

4.3 Objective gap vs. β

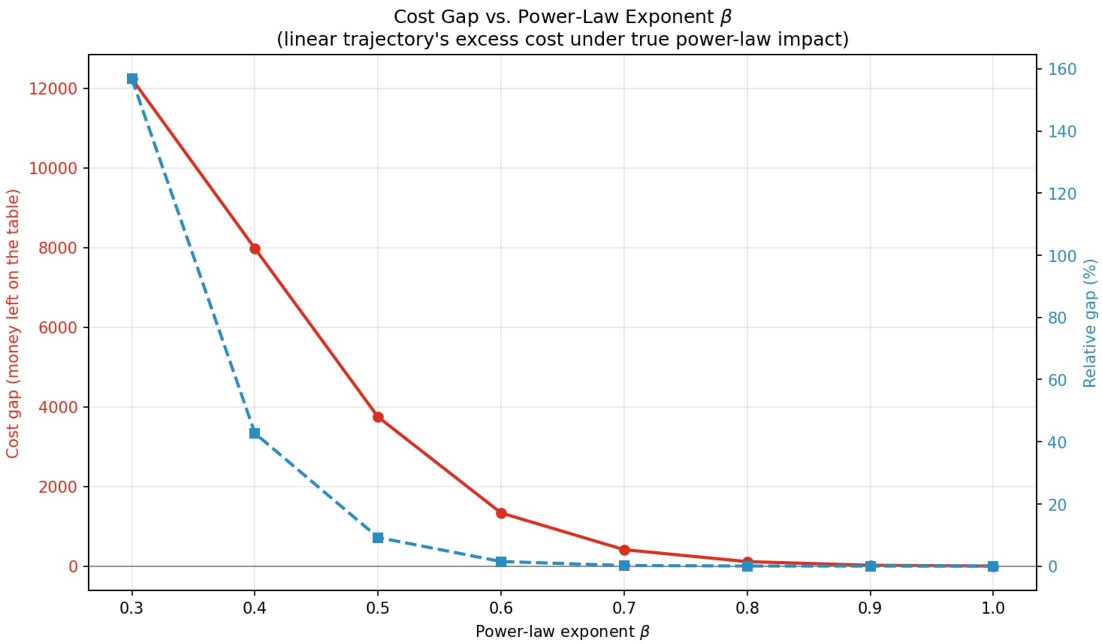

Figure 2. Certainty-equivalent objective gap (excess objective under the power-law model) of the linear trajectory, as a function of the true impact exponent β. The gap grows super-linearly as β decreases.

β	gap (abs)	gap (%)
0.30	$12,239.18	156.8005%
0.40	$7,977.78	42.7291%
0.50	$3,746.11	9.2086%
0.60	$1,335.65	1.4685%
0.70	$412.44	0.1869%
0.80	$111.85	0.0197%
0.90	$20.24	0.0013%
1.00	$0.00	0.0000%

The gap explodes as β drops: 1.47 % at $\beta = 0.6$, 9.2 % at $\beta = 0.5$, 42.7 % at $\beta = 0.4$, and **156.8 %** at $\beta = 0.3$ — the linear strategy's certainty-equivalent cost is more than 2.5× the true optimum. As impact becomes more concave (lower β), large trades are cheaper than the linear model predicts, and the optimum becomes increasingly front-loaded.

4.4 Risk-aversion sensitivity

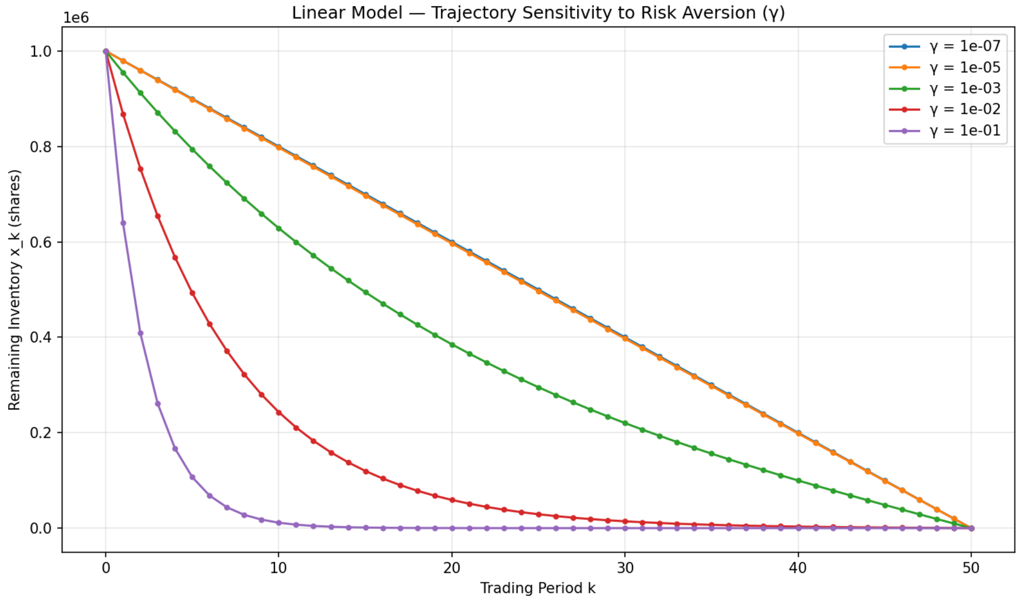

Figure 3. Linear-model trajectories across five risk-aversion levels. Low $\gamma \rightarrow$ TWAP-like; high $\gamma \rightarrow$ sharply convex, front-loaded liquidation.

Risk aversion γ acts as the urgency dial through $\kappa = \sqrt{(\gamma\sigma^2/\eta)}$. The figure shows the expected monotone behavior: higher γ compresses liquidation into the early periods. Combining this with the β-sweep, the linear and power-law models diverge most dramatically at the (γ high, β low) corner — i.e. urgent liquidation in illiquid assets, precisely the regime where execution quality is most financially consequential.

4.5 Cost gap as a function of urgency (κ)

Following feedback from our project review, we extended the sensitivity analysis to study how the cost gap varies with the urgency parameter κ. Recall that $\kappa = \sqrt{(\gamma\sigma^2/\eta)}$ bundles risk aversion, volatility, and impact cost into a single number. A small κ describes a patient liquidation; a large κ describes an urgent one.

We swept κ across nine values from 0.001 (very patient) to 5.0 (extremely urgent), holding σ and η at baseline and adjusting γ accordingly. At each κ we computed both the linear and power-law ($\beta = 0.6$) optimal trajectories, then evaluated the linear trajectory under the power-law cost function. The resulting gap quantifies the certainty-equivalent excess cost of using the linear strategy when reality follows a power-law impact rule.

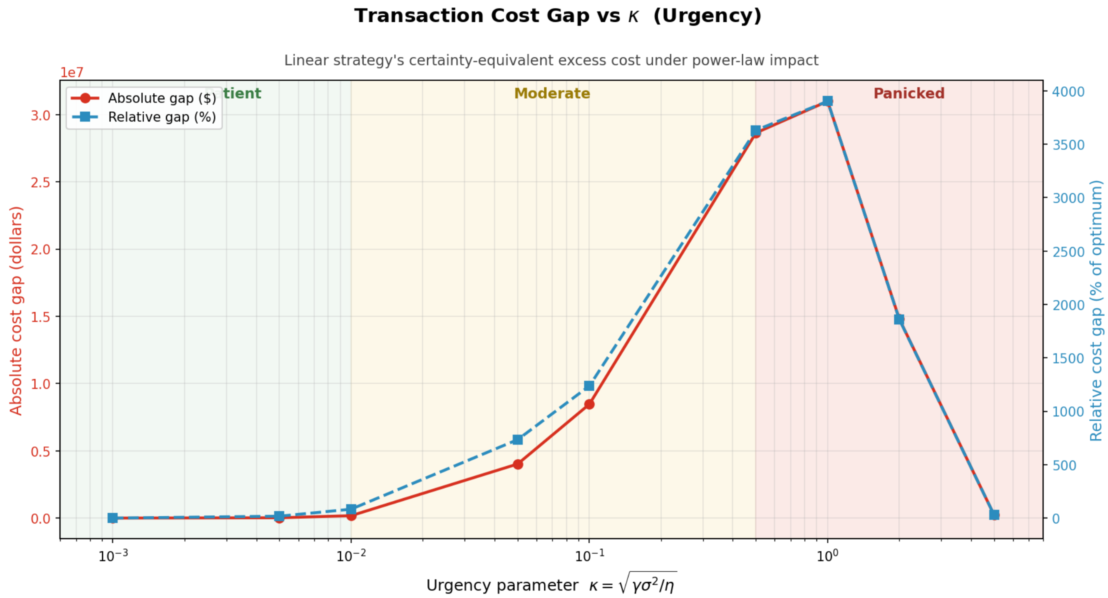

Figure 4. Transaction cost gap as a function of urgency κ. Both absolute (red, left axis) and relative (blue dashed, right axis) representations are shown. The gap is small in the patient regime but grows substantially as κ rises.

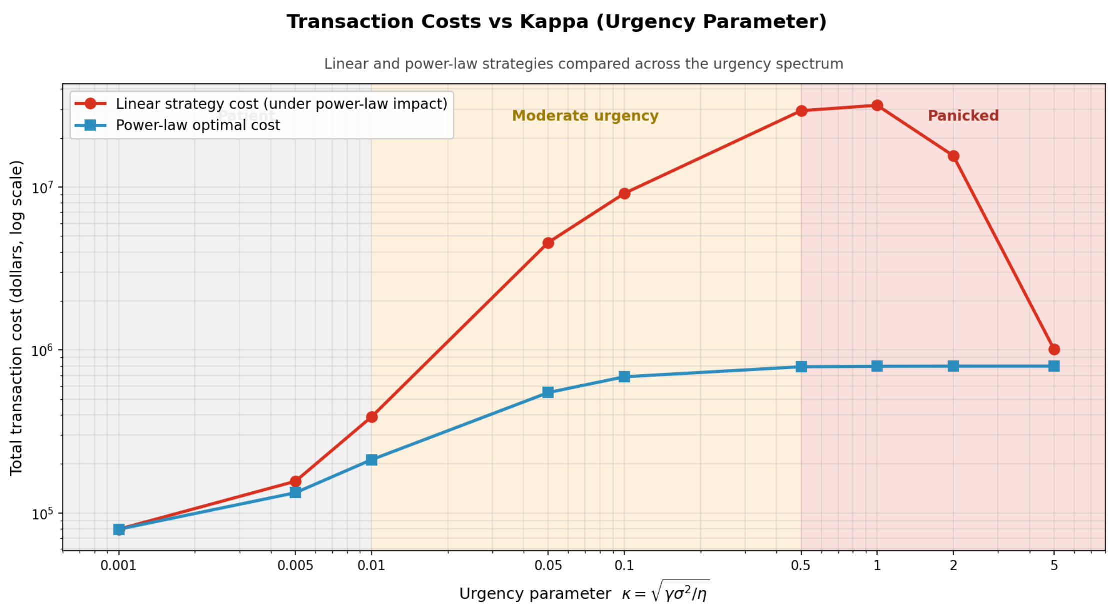

Figure 5. Transaction costs of the linear and power-law strategies plotted across the κ range. The vertical separation between the two lines is the cost gap shown in Figure 4.

Figure 5 presents the same κ sweep in absolute cost terms rather than relative gap. Both strategies' total transaction costs are plotted on a logarithmic axis. The visual divergence makes the practical scale of the disagreement clear: at moderate urgency the linear strategy can cost orders of magnitude more than the power-law optimum. At very high κ, both strategies converge as the optimal action becomes near-immediate liquidation under both impact models.

The chart confirms what we observed in the γ trajectory comparison: when liquidation is patient, both models produce near-identical costs, but as urgency rises the linear model's misspecification of impact becomes increasingly expensive. Combined with the β sensitivity in §4.3, this shows two distinct failure modes of the linear model: it breaks down when impact is highly non-linear (low β) *or* when liquidation is urgent (high κ). In a real execution context, both can occur simultaneously.

We note one technical subtlety: the curve is non-monotone, peaking around $\kappa \approx 0.5$–1.0 and shrinking again at $\kappa = 5$. At extreme urgency, both strategies are forced toward immediate liquidation, so the shapes converge and the relative cost gap contracts.

5. Discussion & Conclusion

At the professor's baseline parameters the linear Almgren-Chriss model carries a certainty-equivalent objective gap of 1.47 % relative to the verified power-law optimum. This is small but not negligible — an order of magnitude larger than the 0.1 % we originally reported using SLSQP alone. The finding that SLSQP was consistently terminating at sub-optimal points is itself a methodological lesson.

The approximation **degrades catastrophically** in two regimes: (i) when the asset exhibits strongly concave impact ($\beta < 0.5$), and (ii) when liquidation is urgent (large γ). In the intersection of these regimes, a linear-impact optimizer can leave triple-digit percentage excess objective on the table. The practical takeaway is that production execution systems should calibrate their impact exponent empirically rather than default to the quadratic form.

5.1 Limitations

Solver robustness. SLSQP hits its iteration cap at extreme parameter values (very low β or very high γ), where the objective is nearly flat near the optimum. This motivated the CVXPY re-solve. A production implementation should use an interior-point solver (CVXPY/SCS/CLARABEL) or the analytical KKT conditions for robustness.

η's implicit units across β. A second methodological caveat concerns the β sensitivity analysis. The coefficient η has implicitly different physical units across different β values, because the impact term $\eta \cdot |v|^{1+\beta}$ has different dimensional scaling. We kept η fixed across β as specified by the project brief, but this means the β sweep mixes two effects: changing impact curvature, and changing impact scale. A fully rigorous sensitivity would renormalize η at each β to match a reference impact cost. We flag this as a limitation rather than a finding.

Future work. Cross-asset impact (portfolio execution), stochastic volatility, and online adaptive execution (reinforcement-learning-based schedules) are natural extensions.

6. References

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